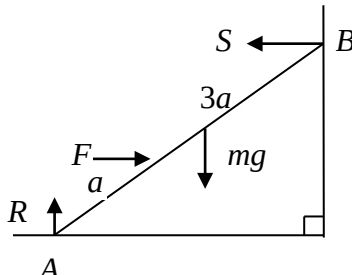
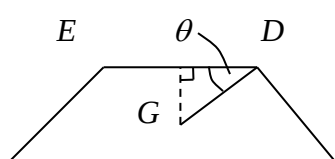
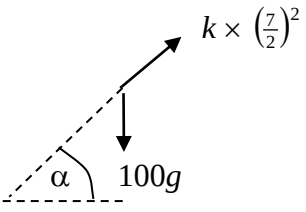
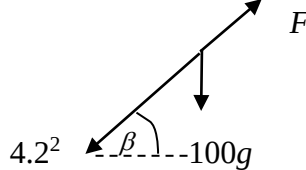
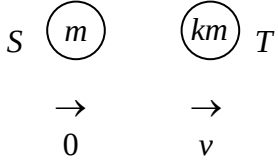


Mock Paper Mark Scheme

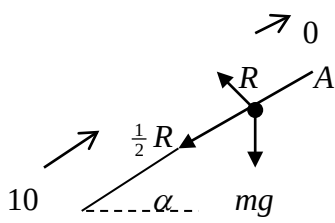
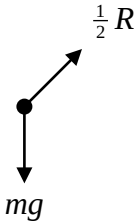
Advanced Subsidiary/Advanced GCE
General Certificate of Education

Subject **MECHANICS**Paper No. **Mock M2**

Question number	Scheme	Marks												
1.	$v = \frac{1}{2}u$ $\text{KE loss} = \frac{1}{2}m(u^2 - (\frac{1}{2}u)^2)$ $= \frac{3mu^2}{8}$ $\therefore \text{fraction of KE lost} = \frac{3mu^2}{8} \div \frac{1}{2}mu^2 = \frac{3}{4}$	B1 M1 A1 M1 A1 (5)												
2.	 $R (\rightarrow), F = S$ M (A) $mg \, 2a \cos \theta + F \, a \sin \theta = S \times 4a \sin \theta$ i.e. $2mg + 2F = 8S$ $F = \frac{1}{3}mg$	B1 M1 A2 M1 A1 (6)												
3. (a)	<table><tr><td></td><td>ABC</td><td>ADE</td><td>BCDE</td></tr><tr><td>Relative mass</td><td>4</td><td>1</td><td>3</td></tr><tr><td>Distance of centre of mass from BC</td><td>10</td><td>20</td><td>$\bar{y}$</td></tr></table> $(4 \times 10) - (1 \times 20) = 3\bar{y}$ $6\frac{2}{3} = \frac{20}{3} = \bar{y} \quad (\text{T})$		ABC	ADE	BCDE	Relative mass	4	1	3	Distance of centre of mass from BC	10	20	$\bar{y}$	B3 (–1 each error or omission) M1 A1 A1 (6)
	ABC	ADE	BCDE											
Relative mass	4	1	3											
Distance of centre of mass from BC	10	20	$\bar{y}$											
(b)	$\tan \theta = \frac{15 - \bar{y}}{20}$ $= \frac{15 - \frac{20}{3}}{20} = \frac{5}{12}$ $\theta = 22.6^\circ \text{ (1 d.p.)}$ 	M1 A1 A1 (3) (9)												
Question number	Scheme	Marks												

4. (a) (b)	 $R (\nearrow), 100g \times \frac{1}{20} = k \times \left(\frac{z}{2}\right)^2$ $\Rightarrow k = 4 \quad (\text{T})$  $R (\nearrow), F - 100g \times \frac{1}{40} - 16 = 0$ $\Rightarrow F = 40.5 \text{ N}$ $P = 40.5 \times 2$ $= 81 \text{ W}$	M1 A1 A1 (3) M1 A2 A1 M1 A1 (6) (9)
5. (a) (b)	$u \rightarrow \quad \rightarrow \lambda u \quad \quad \quad mu + km\lambda u = kmv$  $\frac{u}{k}(1 + k\lambda) = v$ $v = e(u - \lambda u) = eu(1 - \lambda)$ $\Rightarrow \frac{u}{k}(1 + k\lambda) = eu(1 - \lambda)$ $\Rightarrow \frac{(1 + k\lambda)}{k(1 - \lambda)} = e \quad (\text{T})$ $\frac{1 + k\lambda}{k(1 - \lambda)} \leq 1 \quad \Rightarrow 1 + k\lambda \leq k(1 - \lambda)$ $\Rightarrow \frac{1}{1 - 2\lambda} \leq k$ since $0 < \lambda < \frac{1}{2}, k > 1 \quad (\text{T})$	M1 A1 M1 A1 M1 A1 (6) M1 A1 A1 (3) (9)
Question number	Scheme	Marks

6. (a)	$\mathbf{v} = \int 2\mathbf{i} + 6t\mathbf{j} \, dt = 2t\mathbf{i} + 3t^2\mathbf{j} \quad (+ \mathbf{c})$ $\mathbf{c} = 2\mathbf{i} - 4\mathbf{j}$ $\mathbf{v} = (2t + 2)\mathbf{i} + (3t^2 - 4)\mathbf{j}$	M1 A1 A1 (3)
(b)	$t = 2: \mathbf{v} = 6\mathbf{i} + 8\mathbf{j}$ $3\mathbf{i} - 1.5\mathbf{j} = 0.5(\mathbf{v} - (6\mathbf{i} + 8\mathbf{j}))$ $\Rightarrow \mathbf{v} = 12\mathbf{i} + 5\mathbf{j}$ $\Rightarrow \mathbf{v} = \sqrt{(12^2 + 5^2)} = 13 \, \text{m s}^{-1}$	B1 M1 A1 A1 M1 A1 (6) (9)
7. (a)	$(\uparrow): -52.5 = 14t - \frac{1}{2} \times 9.8t^2$ $7t^2 - 20t - 75 = 0$ $(7t + 15)(t - 5) = 0$ $t = 5 \quad (\text{or } t = -\frac{15}{7})$ $(\rightarrow): S = 28 \cos 30^\circ \times 5$ $= 70\sqrt{3} = 121 \, \text{m (3 s.f.)}$	M1 A2 M1 A1 A1 M1 A1 (8)
(b)	$v_{\text{horizontal}} : 28 \cos 30^\circ = 14\sqrt{3}$ $v_{\text{vertical}}: 28 \sin 30^\circ - 5g = -35$ $\therefore \text{speed} = \sqrt{((14\sqrt{3})^2 + 35^2)} = \sqrt{1813} = 42.6 \, \text{m s}^{-1}$	B1 M1 A1 M1 A1 (5) (13)
OR	KE gain = PE loss $\frac{1}{2} m (v^2 - 28^2) = mg \times 52.5$ $\Rightarrow v = \sqrt{1813} = 42.6 \, \text{m s}^{-1}$	M1 A2 M1 A1 (5)

Question number	Scheme	Marks
8. (a)	 $R (\perp), R = mg \cos \alpha = \frac{4}{5} mg$ $\frac{2}{5} mgd = \frac{1}{2} m \times 10^2 - mgd \sin \alpha$ $OA = d = \frac{50}{g} = 5.10 \text{ m (3 s.f.)}$	M1 A1 M1 A3 A1 (7)
(b)	At A ,  component of weight down plane = $mg \sin \alpha = \frac{3mg}{5}$ limiting friction up = $\frac{2mg}{5}$ $\therefore$ slides down as $\frac{3mg}{5} > \frac{2mg}{5}$ Work done against friction = KE loss $2 \times \frac{2mg}{5} \times \frac{50}{g} = \frac{1}{2} m (10^2 - v^2)$ $v = \sqrt{20} = 4.47 \text{ m s}^{-1}$	B1 B1 M1 (3) M1 A3 A1 (5) (15)

